

STRATEGIC VISION: SMART BENGAL CLEANLINESS ECOSYSTEM

Next-Generation Digital Governance & Municipal Accountability Framework

From Complaint to Completion — A Fully Accountable, AI-Powered Municipal Ecosystem

Submitted To: The Hon'ble Minister & Secretary, Urban Development Department, GoWB **Submitted By:**
Roni Paul, Principal Software Architect **Project Codename:** Mission Clean City **Current Status:** Fully
Functional Prototype — Birnagar Municipality, Nadia District **Future Scope:** State-Wide Multi-Municipality
SaaS Platform **Date of Submission:** May 2026

1. EXECUTIVE SUMMARY

Inspired by the visionary **Digital India** mission — launched by the **Hon'ble Prime Minister Shri Narendra Modi** on July 1, 2015 — and in full alignment with the **Smart Bengal** initiative, this proposal presents a revolutionary software ecosystem engineered to eliminate the deep-rooted inefficiencies of traditional urban management. **Urban City Clean** (internally known as Mission Clean City) is not merely a reporting tool; it is a **Centralized Command & Control Center** that empowers municipal leadership with real-time data, citizen-driven oversight, and AI-powered analytics — transforming the way governance is experienced by every resident of Bengal.

By digitizing the entire grievance redressal lifecycle — from the moment a citizen photographs a blocked drain to the moment a supervisor uploads verified proof of its resolution — we ensure that "**Clean Bengal**" is not just a slogan, but a **data-verified, audit-ready reality**. This platform has already been built and tested end-to-end as a fully functional prototype for Birnagar Municipality, Nadia District, and is ready for immediate state-level deployment.

2. THE PROBLEM: WHY TRADITIONAL MUNICIPAL MANAGEMENT IS FAILING

Every municipal body in West Bengal faces the same structural challenges. These are not failures of the people working within the system — they are failures of the **tools** those people have been given.

Challenge	Current Reality	Cost to Government
No Real-Time Visibility	Supervisors learn of problems only after escalation	Delayed response, public frustration
Paper-Based Logs	Complaint registers are lost, illegible, or forged	No audit trail, zero accountability
No Proof of Resolution	Field staff mark tasks complete with no verification	Ghost-billing, wasted labor funds
No Hotspot Analysis	The same drain blocks every monsoon, never permanently fixed	Repeated repair costs, no root-cause fix
Citizen Disconnection	A complaint is filed — and then silence	Loss of public trust in local government
Aadhaar Data Vulnerability	Citizen IDs stored in plain registers	Privacy violation risk, legal liability

3. THE SOLUTION: WHAT MISSION CLEAN CITY DELIVERS

Mission Clean City is a **three-layer digital governance platform** that connects citizens, field staff, and municipal administrators into a single, seamless, accountable system.

Layer 1 — The Citizen Portal (Public-Facing Interface)

Every registered citizen of Birnagar gets a secure, personal dashboard from which they can:

- **File a Complaint** with up to 3 photographs, a video (up to 100 MB), GPS-pinned location, category, and a written description — professionally auto-enhanced by the built-in **AI Writing Assistant**
- **Submit a Suggestion** for civic improvement with ward-specific targeting
- **Track their complaint in real time** — every status change (Pending → In Progress → Resolved) is instantly visible to the citizen with timestamps
- **Rate the resolution** (1–5 stars) after a complaint is marked Resolved — creating a direct citizen satisfaction feedback loop for administration
- **Re-open a resolved complaint** within a 7-day window if the problem recurs, with full justification

- **Appeal a Terminated complaint** — a terminated complaint where the citizen disagrees with the outcome can be formally appealed and re-entered into the review queue
- **Chat with "Birni"** — a 24/7 AI assistant pre-trained on Birnagar's ward structure, municipal departments, and civic processes, answering hundreds of routine citizen queries without requiring any staff time

Security features protecting every citizen:

- Registration is gated by **Email OTP Verification** — no fake accounts
- **One-Aadhaar-One-Account** enforcement via cryptographic SHA-256 hashing — no duplicate registrations
- Aadhaar numbers are stored using **Fernet symmetric encryption at rest** — even a database breach cannot expose a citizen's Aadhaar
- Rate limiting: a citizen can file **2 complaints per day / 10 per month** — preventing spam and system abuse
- **Brute-force login protection**: accounts are locked for 10 minutes after 5 failed attempts
- Profile photos are processed through **AI Face Detection (OpenCV)** — automatically cropped, compressed, and standardized

Layer 2 — The Administrative Command Center

Municipal administrators access a purpose-built dashboard that provides complete oversight of all civic activity across all 14 wards:

Complaint Management:

- View, filter, search, and sort all complaints by ward, status, category, priority, date range, or free-text keyword
- Assign **priority levels** (Low / Medium / High / Urgent) to individual complaints
- Update complaint status with **resolution proof photograph** — a complaint cannot be marked Resolved without a photo uploaded by the admin
- **Bulk actions**: update the status of multiple complaints simultaneously (e.g., mark all garbage complaints in Ward 5 as In Progress at once)
- **Preserve filter state**: the dashboard remembers the admin's filter configuration across actions, eliminating repeated re-filtering
- Generate and print a formal **Digital Work Order** for any complaint — ready for field dispatch

Citizen Directory:

- A full, searchable directory of all registered citizens with ward-level filtering
- View individual citizen profiles including their full complaint and suggestion history
- Securely view **decrypted Aadhaar numbers** — only accessible to authenticated admins, with full access logging

Suggestion Management:

- Review all citizen suggestions with full filtering capabilities
- Accept or reject suggestions with formal status tracking visible to the submitting citizen
- Print formal suggestion review documents

CSV Export:

- One-click export of all complaints or suggestions to structured CSV — ready for state-level reporting to the Urban Development Department

Layer 3 — The Analytics Intelligence Suite

For the Planning & Finance Committee — this is where the platform's governance value becomes quantifiable.

The analytics module transforms raw complaint data into **actionable intelligence**:

- **Real-time KPIs:** Total complaints, resolution rate (%), average citizen satisfaction rating, pending count, in-progress count, terminated count
- **Ward Performance League Table:** Every ward ranked by resolution rate — instantly identifies which wards need more resources or better supervision
- **Category Heatmap:** Which types of complaints (Garbage, Drainage, Roads, Water Supply, Streetlights, Public Health) are most frequent and least resolved
- **Trend Analysis:** Complaint submission and resolution trends over today / 7 days / 30 days / 90 days / 6 months / 1 year — or any custom date range
- **Citizen Growth Chart:** Track platform adoption over time — a direct indicator of public trust
- **Top Problematic Ward Alert:** Instantly surfaced — the ward with the highest unresolved complaint count is always visible at a glance
- **Highest-Impact Category:** The complaint category with the most filings — for targeted resource allocation
- All charts rendered with **1-hour intelligent caching** — the dashboard loads instantly even with thousands of records

This is not reporting after the fact. This is continuous, real-time situational awareness for municipal leadership.

4. PLATFORM SCREENSHOTS — FUNCTIONAL PROTOTYPE

4.1 Citizen Interface

1. Citizen Secure Login

The screenshot displays the Birnagar Municipality Citizen Login interface. At the top left, the logo for Birnagar Municipality is shown with the tagline "MISSION CLEAN CITY". To the right of the logo, navigation links for "Home", "About", "Suggestions", and "Login" are visible, along with a "Report Issue" button. The main heading "Citizen Login" is centered, with a sub-link "Or create a new account" below it. The login form is a white box containing a "Mobile Number" field with "988" entered, a "Password" field with masked characters, a "Forgot your password?" link, and a "Sign in" button. A small chatbot icon is located in the bottom right corner.

The citizen login portal of Birnagar Municipality. Residents sign in using their registered mobile number and password. The "Birni" AI chatbot (bottom right) is available at all times for civic assistance — even before logging in.

2. Citizen Personal Dashboard

Birnagar Municipality
MISSION CLEAN CITY

Home About Suggestions Submit Issue **My Dashboard** Hi, Roni Paul [Logout](#)

Roni Paul
ID: BM-199817

MOBILE NUMBER: 988
AADHAAR NUMBER: XXXX-XXXX-1704
WARD NUMBER: 5
TOTAL REPORTS: 7

Track an Issue

Enter Problem ID (e.g., BIR-WARD4-1029) [Search](#)

My Submission History

Issues Reported **7** Submitted Suggestions **1**

Status: All Statuses
Category: All Categories

After login, citizens see their full profile — Citizen ID (BM-XXXXXX), masked Aadhaar, ward number, and total reports filed. The "Track an Issue" search bar lets citizens look up any complaint by its unique Problem ID. The submission history tab displays all past complaints and suggestions with live status updates.

3. Multimedia Complaint Reporting Engine



Birnagar Municipality
MISSION CLEAN CITY

[Home](#) [About](#) [Suggestions](#) [Submit Issue](#) [My Dashboard](#)

 Hi, Roni Paul [Logout](#)

Report a Problem

Your Ward: 5

Submission Limits

You can submit max 2 reports daily and 10 per month.

Daily Left: 2/2

Monthly Left: 10/10

Photos of the Issue (Max 3, Min 1) *

JPEG/JPG images only. Photos will be auto-compressed to under 75KB.

Upload Photo 1 *

Upload Photo 2

Upload Photo 3

Video Proof (Optional, Max 100MB)

Upload Video (MP4, AVI, etc.)

Problem Category *

Select a Problem Category

Pin Exact Location (Required)

Click on the map to place a pin at the exact location of the problem. Address will be fetched automatically.



No location selected.

Clear Pin

Fetch Address *

Address will auto-fill when you pin the location

Submit Issue

The "Report a Problem" interface shows the citizen's ward, daily and monthly submission limits (anti-spam enforcement), photo upload slots (up to 3, auto-compressed to under 75 KB), optional video proof, problem category selector, and an embedded GIS map where the citizen pins the exact location. The address is auto-fetched from the GPS coordinates — no manual typing required.

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4. Civic Suggestion Portal



Birnagar Municipality
MISSION CLEAN CITY

[Home](#) [About](#) [Suggestions](#) [Submit Issue](#) [My Dashboard](#)

 [Hi, Roni Paul](#) [Logout](#)

Suggestions Box

Have an idea on what we can improve for our city? We're all ears.

Submission Limits
You can submit max 1 suggestion daily and 5 per month.

Daily Left: 1/1
Monthly Left: 5/5

Your Information

Full Name
Roni Paul

Mobile Number
988

Aadhaar Number
XXXX-XXXX-1704

Your Ward Number
Ward 5

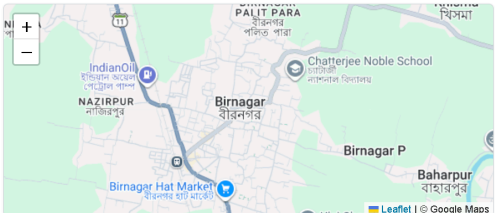
Your Address
Birnagar, Nadia, West Bengal, India - Updated

Improvement Area

Target Ward Number *
Select Target Ward

Target Address *
Full address of the area that needs improvement

Pin Exact Location (Optional but recommended)
Click on the map to place a pin at the exact location that needs improvement.



No location selected. [Clear Pin](#)

Suggestion Details

Suggestion Category *
Select a Category

Your Idea / Suggestion
Tell us how we can make Birnagar better...

Photo of this place *

[Choose File](#) No file chosen

JPEG/JPG Images only, Auto-compressed to under 75KB.

[Submit Suggestion](#)

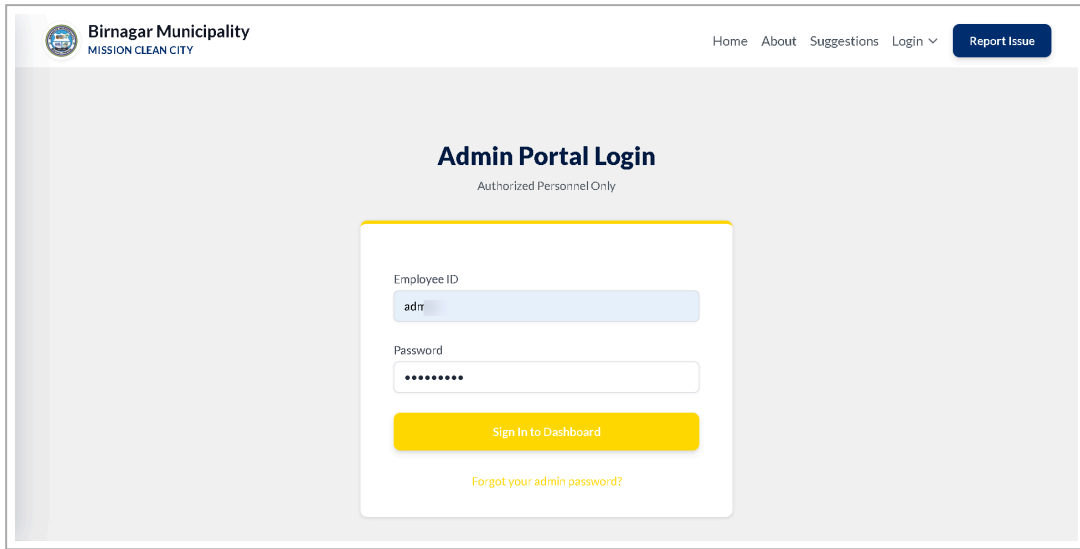
Submissions are reviewed weekly by the municipal board.

The Suggestions Box allows citizens to submit structured improvement ideas to the municipality. The form captures the citizen's profile details automatically, the target ward and address, a GIS location pin, a category (Water, Roads, Parks, Drainage, etc.), a written description, and a supporting photograph. Submissions are reviewed weekly by the municipal board.

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4.2 Administrative Command Center

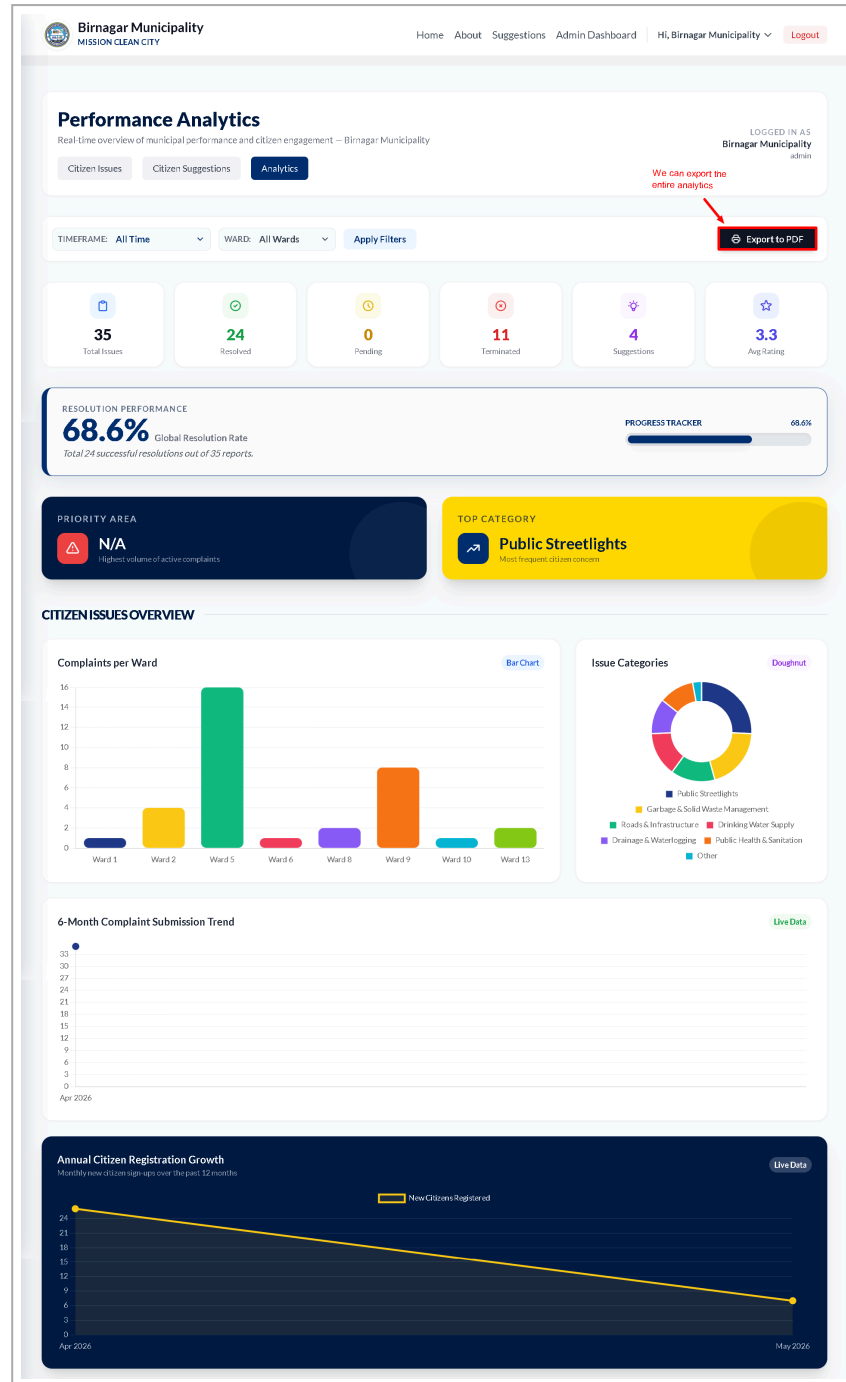
5. Admin Secure Gateway



The screenshot shows the 'Admin Portal Login' page for Birnagar Municipality. The header includes the municipality's logo and name, a navigation menu with 'Home', 'About', 'Suggestions', and 'Login', and a 'Report Issue' button. The main content area features a central login form with a yellow border. The form has a title 'Admin Portal Login' and a subtitle 'Authorized Personnel Only'. It contains two input fields: 'Employee ID' with the text 'adm' and 'Password' with masked characters. Below the fields is a yellow 'Sign In to Dashboard' button and a link 'Forgot your admin password?'.

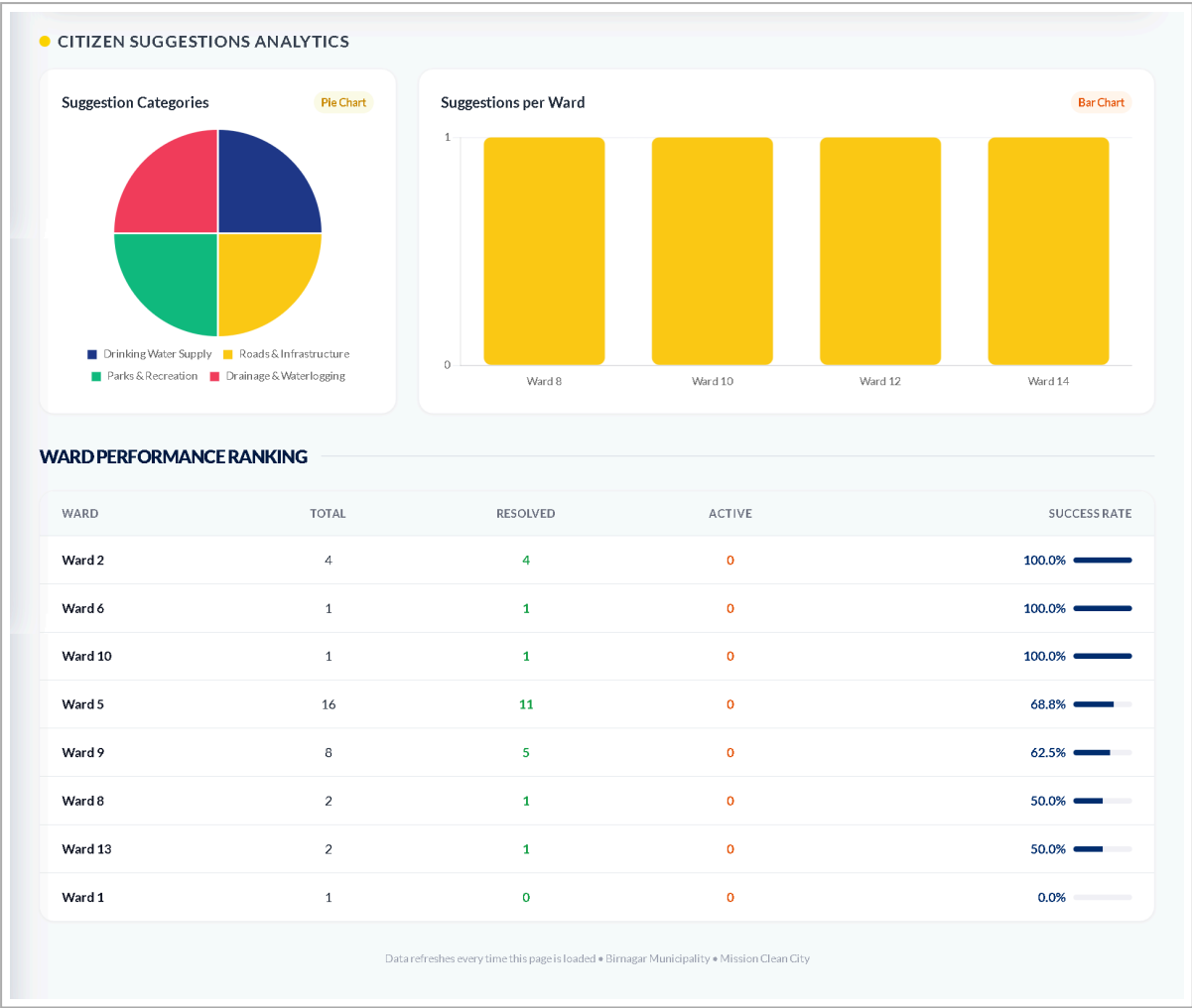
The Admin Portal Login is visually distinct from the citizen interface — clearly marked "Authorized Personnel Only." Admins authenticate via their Employee ID and password. The yellow-accent design separates it unmistakably from the public-facing portal, with a 15-minute brute-force lockout protecting admin accounts.

6. Executive Analytics Dashboard — Citizen Issues Overview



The Performance Analytics dashboard displays a complete municipal intelligence overview. Live KPIs at the top show 35 total issues, 24 resolved, 0 pending, 11 terminated, 4 suggestions, and a 3.3 average citizen satisfaction rating — with a global resolution rate of 68.6% visualised on a progress bar. The Priority Area panel flags the ward with the highest volume of active complaints, while the Top Category card highlights "Public Streetlights" as the most frequent citizen concern. Below, the Citizen Issues Overview section presents ward-wise complaint density as a bar chart, issue category distribution as a donut chart (covering Streetlights, Garbage, Roads, Drinking Water, Drainage, and Public Health), a 6-month complaint submission trend line, and an Annual Citizen Registration Growth chart tracking new sign-ups month by month. All data is filterable by timeframe and ward and can be exported to PDF with one click.

6b. Executive Analytics Dashboard — Suggestions & Ward Performance



The lower section of the analytics dashboard presents *Citizen Suggestions Analytics* — a pie chart breaking down suggestion categories (*Drinking Water Supply*, *Roads & Infrastructure*, *Parks & Recreation*, *Drainage & Waterlogging*) and a bar chart showing suggestions per ward (*Wards 8, 10, 12, and 14*). Below this, the *Ward Performance Ranking* table ranks all active wards by success rate: *Ward 2, Ward 6, and Ward 10* each achieve 100.0% resolution; *Ward 5* — the highest-volume ward with 16 total complaints — achieves 68.8%; *Ward 9* at 62.5%; and *Ward 1* at 0.0%, immediately flagging it for administrative intervention. This table gives municipal leadership instant, data-backed visibility into where resources must be directed.

7. Complaint Command Center



Birnagar Municipality

MISSION CLEAN CITY

Home

About

Suggestions

Admin Dashboard

Hi, Birnagar Municipality

Logout

Admin Dashboard

Welcome, Birnagar Municipality (ID: admin)

Search ID, Name, Mobile, Description, Address...

All Wards

All Statuses

All Categories

All Priorities

From: dd-mm-yyyy

To: dd-mm-yyyy

Search

QUICK FILTERS:

Urgent & Pending

In Progress

Unrated Resolved

Appealed

Today's Issues

Pending

0

In Progress

0

Solved

24

Terminated

11

Citizen Issues

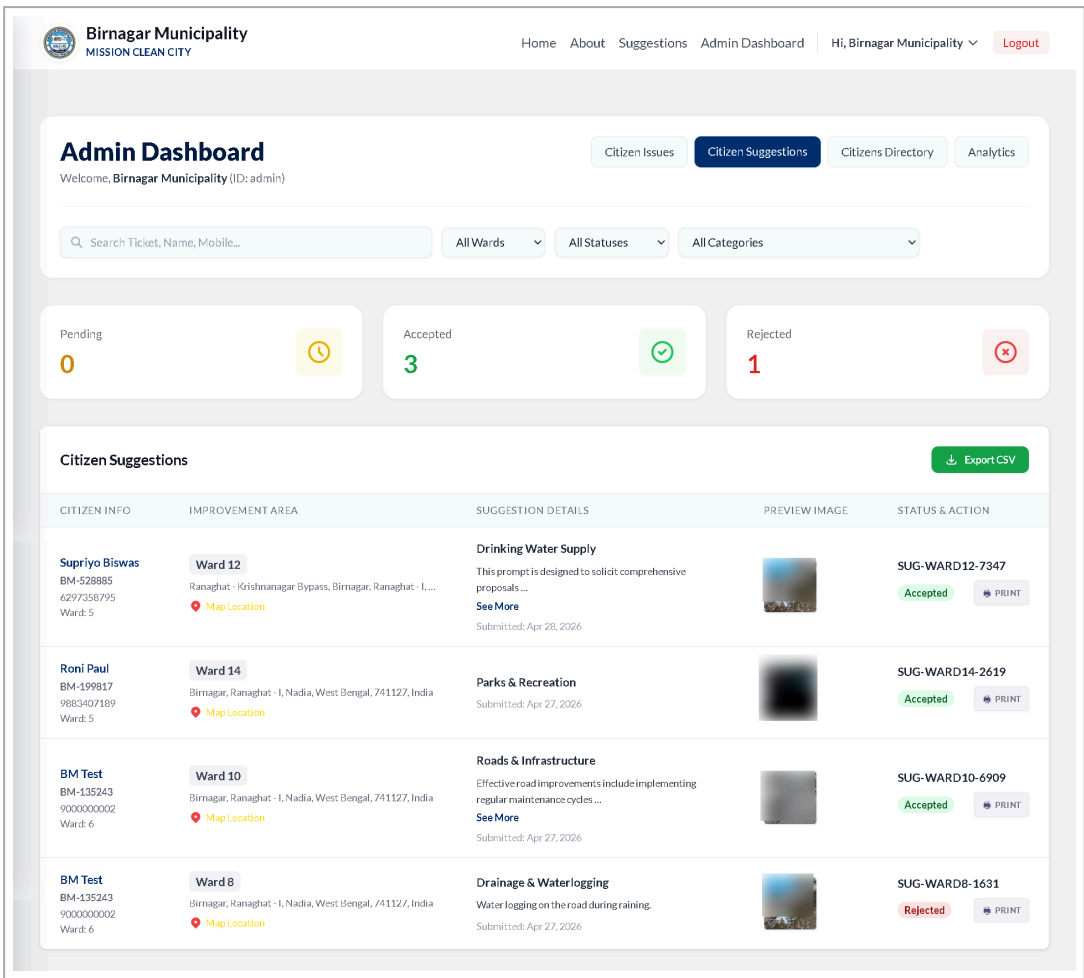
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Export CSV

	CITIZEN INFO	LOCATION & ID	DETAILS	PROOF	RATING	ACTION & STATUS
	Hello D BM-120539 9313845976 Ward: 2	BIR-WARD2-8579 Birnagar, Ranaghat - I, Nadia, West Bengal, 741127... Map Location	Public Streetlights Streetlight pole is physically damaged, tilted, or unsafe Appeal Reason: Please look at this matter more carefully. SUBMITTED: APR 30, 2026		★★★★★ Rated	Resolved Done. Status Locked PRINT
	Roni Paul BM-199817 9883407189 Ward: 5	BIR-WARD5-7745 Birnagar, Ranaghat - I, Nadia, West Bengal, 741127... Map Location	Drinking Water Supply Appeal Reason: Please take it back. SUBMITTED: APR 30, 2026		Not Rated	Terminated Already done with the photo. Status Locked PRINT
	Roni Paul BM-199817 9883407189 Ward: 5	BIR-WARD5-4630 Birnagar, Ranaghat - I, Nadia, West Bengal, 741127... Map Location	Drinking Water Supply SUBMITTED: APR 30, 2026		★★★★★ Rated	Resolved Great Work. Status Locked PRINT

The Admin Dashboard's Citizen Issues view gives administrators complete control over the complaint pipeline. Quick filters surface Urgent & Pending, In Progress, Unrated Resolved, Appealed, and Today's Issues with one click. Each complaint row shows the citizen's full profile, unique Problem ID (BIR-WARD#-XXXX), GPS map link, issue category and description, resolution proof photo thumbnail, citizen star rating, and action controls — including a one-click Print Work Order button for field dispatch.

8. Citizen Suggestion Review Hub



The Citizen Suggestions management panel allows administrators to review all public improvement submissions. Each entry displays the submitting citizen's details, the target ward with a map location link, the suggestion category and description, a supporting photo preview, and a unique Suggestion Ticket ID (SUG-WARD#-XXXX). Admins can Accept or Reject suggestions — and the citizen's dashboard reflects the status immediately. All suggestion data can be exported to CSV for policy planning.

5. TECHNICAL ARCHITECTURE & SECURITY POSTURE

Our architecture is designed for **Zero-Trust Environments**.

- **State-of-the-Art Encryption:** Citizen Aadhaar data is protected via **Advanced Symmetric Encryption (Fernet)**. We store only encrypted hashes to ensure privacy — even a database breach cannot expose a citizen's identity.
- **Multi-Factor Integrity:** All critical actions — password resets, email changes, new registrations — are protected by **6-digit cryptographic OTPs** delivered via secure SMTP and stored only as one-way SHA-256 hashes. The raw OTP never persists anywhere in the system.
- **Infrastructure:** Python/Django backend with a **PostgreSQL-ready schema** for massive concurrent traffic. SQLite WAL journal mode enabled in development for concurrent read/write performance. WSGI/ASGI-compatible — deployable on NIC Cloud, AWS, or any government-grade VPS.

- **AI Integration:** Powered by **Google Gemini NLP**, providing a 24/7 intelligent municipal guide for citizens, with a Bytez API fallback ensuring 100% AI uptime.
- **Session Security:** HttpOnly + Secure cookies, CSRF protection on all state-changing requests, HSTS enforced in production, and X-Frame-Options: DENY to prevent clickjacking.

6. ECONOMIC IMPACT — WHY APPROVAL IS A FISCAL NECESSITY

For the Finance Department and Planning Commission.

This platform is not a cost. It is a **cost-reduction mechanism with measurable, auditable outcomes.**

6.1 Direct Cost Savings

Area	Current Waste	Platform Solution	Estimated Annual Saving
Fuel & Fleet	Garbage trucks run blind fixed routes daily	GIS heatmaps deploy vehicles <i>only</i> where complaints are active	30–40% reduction in fuel costs
Labor Billing	No verification that field work was actually done	Photo-proof mandatory for every resolved complaint — no photo, no resolution	Eliminates ghost-billing entirely
Administrative Staff Time	Manual data entry, phone calls, paper-based filing	Zero-paper, self-service citizen reporting — automated end to end	Hundreds of staff-hours recovered per month
Repeat Infrastructure Repairs	Same drain repaired repeatedly with no root-cause fix	Trend analytics pinpoint chronic problem locations for permanent structural repair	Shifts spending from reactive repair to prevention

6.2 Optimizing Taxpayer Funds & Budgeting

- **Predictive Maintenance:** Analytics identify chronic infrastructure failures (e.g., the same drain blocking repeatedly). Instead of wasting funds on weekly temporary fixes, resources can be allocated for permanent structural repair.
- **Data-Driven Budgeting:** Identify exactly where funds are needed based on infrastructure failure heatmaps — budget deployed with surgical precision rather than guesswork.
- **Zero-Paper Administration:** Eliminating physical ledgers, manual data entry, and fragmented communication saves thousands of administrative hours per month.

6.3 Strategic & Political Alignment

- **Nirmal Bangla Vision:** Directly supports the mandate of a clean Bengal, providing **verifiable, digital metrics** to showcase the government's absolute commitment to public health and sanitation before the Ministry of Housing & Urban Affairs.

- **Grant Eligibility:** The platform's architecture is aligned with Smart Cities Mission and AMRUT 2.0 eligibility criteria — digital governance platforms often unlock significant central government grants.

7. FUTURE SCOPE — THE "NIRMAL WEST BENGAL" MULTI-MUNICIPALITY PLATFORM

We have engineered this platform to be a **State-Wide Multi-Tenant Infrastructure**. The Birnagar prototype is Phase 1 — the architecture was designed from day one to scale to every municipality in West Bengal.

- **Municipality-as-a-Service (MaaS):** A single cloud instance can host **100+ municipalities simultaneously** with 100% data isolation. Each municipality gets its own branded portal, while the state government receives a unified super-admin dashboard with aggregate analytics across all municipalities in one view.
- **IoT Ready:** Future integration for "**Smart Sensors**" in dustbins to automate pickup schedules — bins signal full capacity directly to the municipal dashboard, eliminating manual inspection rounds entirely.
- **Native Mobility:** Development of native **iOS and Android apps** for field staff and citizens is underway — enabling complaint filing, work order receipt, and resolution proof upload directly from smartphones in the field.

8. CONCLUSION

Urban City Clean is the definitive answer to the challenges of modern urban sanitation in West Bengal. It replaces doubt with data, and complaints with completion. By approving this proposal, the Urban Development Department will not just be enabling software — it will be implementing a **Gold Standard of Governance** that other states will look to as a model of excellence; a living proof that "Nirmal Bangla" is not an aspiration, but a measurable, daily-verified reality.

We are ready to scale. We are ready to clean. We are ready for a Smart West Bengal.

This document is submitted in good faith to the Urban Development & Municipal Affairs Department, Government of WB.
All platform features described herein are implemented and verifiable in the working prototype.

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